

Discrete Regimes, Carefully: Mode Uncertainty in the Obstruction Calculus

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Abstract

Discrete regimes are where hybrid formalisms tempt and where decorative extensions hide. This paper, the D11 item of the generalization catalogue, takes the catalogue’s own warning seriously — hybrid structure only where the mode is load-bearing — and exhibits the smallest setting in which it is. The working system gains an exogenous regime $m \in \{1, 2\}$ chosen adversarially each step: each patch regenerates, and the pressured patch takes damage unless protected. Under either *frozen* regime every safe pair is viable (36 of 36; each regime is separately sustainable); with the regime *unread*, the mode-blind kernel holds exactly 28 of 36 — eight beliefs, precisely those containing the thin-stock corner $(1, 1)$, are lost purely to unread regimes; with the regime *read* per (state, mode) cell, protecting the pressured patch restores all 36. The chain $\text{Viab}_{\text{blind}} \subset \text{Viab}_{\text{observed}} = \text{Viab}_{\text{frozen-1}} \cap \text{Viab}_{\text{frozen-2}}$ is exact, and the recursive augmented-belief computation is cross-validated against brute-force game enumeration. Scope is recorded, not claimed: finite modes, total transitions, no Zeno, no continuous-mode metrics (6/6 check families).

Introduction and relation to the catalogue’s warning

The catalogue proposes hybrid generalization “carefully”: only if the mode structure is central, otherwise decorative. The paper honors this by construction: the mode is load-bearing exactly when it is adversarial and unread — Section 3’s eight lost beliefs vanish under either frozen regime and under full reading, so the obstruction belongs to mode *uncertainty* alone. The paper’s model class is finite discrete modes with total transitions; the catalogue’s warning against decorative hybrid machinery (Zeno, timed-automata metrics, continuous mode dynamics) is adopted verbatim as the scope statement.

Mode uncertainty

Branches carry (x, m) ; the per-cell read-then-act semantics applies on augmented cells; after the regulator’s per-cell move the adversary re-picks the mode. Three policy classes: frozen mode (no switching), mode-blind (cells

keyed by state), mode-observed (cells keyed by (x, m)).

Proposition 1 (Frozen regimes are separately sustainable). With the mode frozen at either value, protecting the pressured patch lets both patches grow: every safe pair is viable under each frozen regime (36 of 36).

Proposition 2 (The mode-uncertainty obstruction and its exact boundary). Under an unread adversarial regime the mode-blind kernel holds exactly 28 of 36 safe pairs; the eight lost beliefs are exactly those containing $(1, 1)$; under mode observation, protecting the pressured patch restores all 36. Hence $\text{Viab}_{\text{blind}} \subset \text{Viab}_{\text{observed}} = \text{Viab}_{\text{frozen-1}} \cap \text{Viab}_{\text{frozen-2}}$ setwise.

Proof. The counts, the lost-family characterization (all containing $(1, 1)$), and the setwise chain are computed exactly by the verification script; the restoration strategy is $u = m$ per observed cell. The boundary reading: thin-buffer beliefs cannot juggle an unread adversary, buffered ones can. $\square$

Verified record (6/6 exact check families)

E1 frozen-mode kernels $36 = 36$; E2 the mode-blind kernel exactly 28 of 36 with the eight lost beliefs enumerated (all contain $(1, 1)$); E3 mode-observed restoration 36 of 36; E4 cross-validation of the recursive augmented computation against brute-force game-tree enumeration (three probe beliefs $\times$ horizons 1–4 $\times$ frozen/adversarial switching); E5 the exact policy-class chain ($28 \subset 36 = 36$); E6 the scope assertions (finite modes, total transitions, load-bearing instance exhibited).

What is not claimed

No general hybrid-systems theory: no Zeno analysis, no timed or metric mode semantics, no stochastic mode processes, no continuous-mode dynamics. The regime is adversarial and exogenous; endogenous regime change (institutional responses to stock levels) is cited as the governance extension and not developed. The damage-regeneration asymmetry is the audited instance’s; no classification over damage profiles is claimed. The paper’s honesty obligation is the catalogue’s warning itself: where

the mode is not load-bearing (frozen or observed), the hybrid layer disappears — and the paper says so.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `hybrid_mode_viability_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.